



SYSTEMS ENGINEERING / RUNTIME RECOVERY

When workflows fail, state still matters.

A SPARTAN 337 reliability and recovery design sample showing explicit health checks, evidence validity, degraded-state behavior, checkpoint protection, restart reconciliation, and operator-visible recovery paths.

RUNTIME CONTINUITY

Fail safe. Reconcile. Resume.

-  **HEALTHY**
accept work
-  **DEGRADED**
protect state
-  **RECOVERING**
reconcile checkpoint
-  **READY**
resume governed queue

DESIGN RULE

Invalid evidence blocks confident action;
restart never implies hidden completion.

PERSONAL SYSTEMS ENGINEERING PROJECT / RECOVERY DEMONSTRATION

PUBLIC-FACING, SANITIZED PROJECT DEMONSTRATION

FAILURE DETECTION

DEGRADED MODE

CHECKPOINT RECOVERY

SAFE RESUME





Runtime Reliability & Recovery Design

PROBLEM

A workflow is only useful if it fails predictably. Crashes, stale evidence, checkpoint drift, or queue mismatches can create false completion or repeat work unless recovery behavior is explicit.

APPROACH

Detect invariant failures, degrade safely, preserve durable state, reconcile checkpoint and evidence, repair only what is necessary, and resume from a verified state while exposing the reason and next safe action to the operator.

Result: failure handling is explicit, state-aware, and visible to the operator.

This is a recovery design / acceptance model, not a production uptime or live telemetry claim.

RECOVERY SEQUENCE

- 1 Detect**
Health check / invariant fails
- 2 Degrade**
Stop confident or side-effecting action
- 3 Preserve**
Keep durable queue + checkpoint intact
- 4 Reconcile**
Compare checkpoint, evidence, and state
- 5 Resume**
Continue only from a verified state

FAILURE / RESPONSE MATRIX

Evidence invalid Block decision	WAIT_EVIDENCE
Heartbeat missing Degraded mode	DEGRADED
Checkpoint stale Reconcile state	RECOVERING
Queue mismatch Protect + repair	RECOVERING
Health restored Verify then resume	READY

OPERATOR VISIBILITY: state + reason + next safe action

CAPABILITIES DEMONSTRATED

Reliability Engineering / Failure Analysis / State Recovery / Checkpointing / Operational QA / System Observability

